Oscar I. Hernandez

Summary

A graduate student of mathematics, I'm looking for a summer research internship.

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Citizenship: USA

Education

University of Alaska Fairbanks (UAF CNSM DMS)
M.S., Mathematics — Aug 2020 –

Bard College at Simon's Rock (BCSR)
B.A., Mathematics and Computer Science, Honors Thesis [HW17] — Aug 2013 – May 2017
A.A., Liberal Arts and Sciences — May 2015

Awards: W.E.B. Du Bois Scholarship (2013–2017), Alumni Scholarship (2013–2017), Supplemental Return Grant (2014–2017), Dean's List (Falls 2015 & 2016).

Experience

Generalist Engineer	Independent Freelancer	Jan 2018 – July 2020
Founder & Robotics Engineer	ARKS	Apr 2018 – Mar 2020
Research Intern/Assistant	BCSR: Comp Sci, Econ	May 2015 – May 2016

Teaching

Graduate Teaching Assistant	UAF CNSM DMS	Aug 2020 – May 2021
Calculus I		
5th Grade Humanities Teacher	SACS HNW	Jul 2017 – Decr 2017
G5 World History	G5 Math	G5–6 CS Quarter Elective
G5 English Language Arts	G5 Earth Science	G5–8 CS Semester Elective
Subject/Class Tutor	BCSR	Aug 2014 – May 2017
Computer Science	Latin I	Intro to Computer Science
Critical Reasoning	Formal Logic	Algorithms & Data Structures
Algebra I with Geometry	Discrete Math	Graph Theory & Combinatorics
Algebra II and Trigonometry	Vector Calculus	

Awards

1. NSF Innovation Corps grant recipient x2 (UCR EPIC Spring 2019, CSUN Fall 2019)

Service

1. Reviewer (ad hoc) for the *Journal of Information Processing – Special Issue of "Discrete and Computational Geometry, Graphs, and Games"*

Publications

Journal publication (1), thesis (1), and conference proceedings (2):

[Her+18] Oscar I. Hernandez et al. "Constructing de Bruijn sequences with co-lexicographic order: The k-ary Grandmama sequence". In: *European Journal of Combinatorics* 72 (2018), pp. 1–11. ISSN: 0195-6698. URL: http://www.sciencedirect.com/science/article/pii/S0195669818300696.

[Has+17] Harold M. Hastings et al. "Dynamics of Biomimetic Electronic Artificial Neural Networks". In: *Proceedings of the 4th International Conference on Applications in Nonlinear Dynamics (ICAND 2016)*. Ed. by Visarath In, Patrick Longhini, and Antonio Palacios. Cham: Springer International Publishing, 2017, pp. 195–207. ISBN: 978-3-319-52621-8. URL: https://link.springer.com/chapter/10.1007%5C%2F978-3-319-52621-8_18.

[HW17] Oscar I Hernandez and Aaron Williams. "Constructing Maximum-Rotating de Bruijn Sequences using Necklace Trees". *Bachelor of Arts* Senior Thesis awarded Honors. Bard College at Simon's Rock: Division of Science, Mathematics, and Computing, May 2017. URL: https://digitalcommons.bard.edu/sr-theses/1120.

[DHW16] Patrick Baxter Dragon, Oscar I. Hernandez, and Aaron Williams. "The Grandmama de Bruijn Sequence for Binary Strings". In: *LATIN 2016: Theoretical Informatics*. Ed. by Evangelos Kranakis, Gonzalo Navarro, and Edgar Chávez. Berlin, Heidelberg: Springer Berlin Heidelberg, 2016, pp. 347–361. ISBN: 978-3-662-49529-2. URL: https://link.springer.com/chapter/10.1007/978-3-662-49529-2_26.

Workshop papers (2), invited presentations (2), and small articles (1):

[HY20] *Lower Gauge Theory*. Special Interest Group on Harry Q Bovik. Similar joke articles published annually since 2016. Association for Computational Heresy. Carnegie Mellon University, Apr. 2020. URL: http://sigbovik.org/2020/proceedings.pdf#page=91.

[HI19] *Zero Shot Sentiment Analysis on Tweets in Any Language*. Poster Session #1. Southern California Natural Language Processing Symposium. University of Southern California, Sept. 2019. URL: https://socalnlp.github.io/symp19/index.html.

[HT19] *Toward Compositional Transfer Learning*. Contributed Paper Session 1. American Mathematical Society Fall Western Sectional Meeting. UC Riverside, Nov. 2019. URL: http://www.ams.org/amsmtgs/2266_abstracts/1153-18-420.pdf.

[Her+19] *Spinning Around and Around: Turnstile Puzzles are* `PSPACE`*-Complete*. Work under revision started at the invite-only 32nd Bellairs Winter Workshop on Computational Geometry. 2019.

[Her+16] *Analog Artificial Neural Networks*. Poster Session. Awarded 2nd place and $200 https://www.aps.org/units/nyss/governance/minutes/upload/Fall16.pdf#page=4. New York State Section of the American Physical Society (NYSS-APS) Symposium on the Physics of 2D Materials. Rensselaer Polytechnic Institute, Nov. 2016. URL: https://simons-rock.edu/news/students-win-prize-physics-research.php.